COM668 COMPUTING PROJECT

AT1 Project Proposal - Supervisor Feedback

Student Name: Sanju Baraili
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Project Title: AI Skincare Progress Tracker
Supervisor/Mentor: Oguzie Goodluck
Date of Review: February 12, 2026
Proposal Status: Approved with Minor Recommendations

EXECUTIVE SUMMARY

Sanju has submitted an **excellent project proposal** that demonstrates strong research, clear technical planning, and appropriate academic rigor. The proposal addresses a genuine real-world problem with a well-defined technical solution combining computer vision, machine learning, and software engineering principles.

The project shows: - Comprehensive background research with 10 academic references - Clear, measurable aim and objectives - Strong technical foundation with appropriate technology choices - Realistic scope for a 400-hour Level 6 project - Awareness of ethical considerations and IPR issues - Excellent alignment with BCS guidelines and module learning outcomes

Current Assessment: This proposal meets the required standard for approval. With the minor enhancements suggested below, this project has strong potential to achieve an **Excellent grade (70-85% range)**.

Proposal Status: APPROVED - Subject to addressing ethics approval requirements and minor recommendations outlined below.

DETAILED FEEDBACK BY SECTION

1. PROJECT AIM EXCELLENT

Submission: “To design and develop an AI-powered camera-based visual tracking system that enables consumers to objectively monitor skincare product effectiveness by automatically capturing, aligning, and analysing facial images over time, detecting measurable improvements in acne count, redness intensity, and skin texture, providing quantified progress metrics to eliminate subjective judgment.”

Feedback: This is an **outstanding aim statement**. It is: - **Clear and focused:** Specific technical solution to a defined problem - **Measurable:** Includes quantifiable metrics (acne count, redness intensity, texture) - **Achievable:** Realistic scope for the available timeframe - **Relevant:** Addresses a genuine consumer need - **Specific:** Details the technical approach and expected outcomes

Strengths: - Explicitly states the technology approach (AI-powered, camera-based) - Defines target users (consumers) - Specifies what will be measured (acne, redness, texture) - Clear value proposition (objective vs. subjective monitoring)

Minor Suggestion: Consider slightly shortening for readability (currently 60+ words), but this is a minor point. The content is excellent.

Grade Potential for this section: 75-85%

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2. PROBLEM DESCRIPTION VERY GOOD

Strengths: - **Quantified market context:** £100 billion industry figure adds credibility - **Clear problem identification:** Lack of objective measurement methods - **User impact articulated:** Wasted money, premature abandonment - **Competitive analysis:** Identifies gaps in existing solutions - **Cost comparison:** Professional tools £300-800 vs. affordable consumer solution - **Market need justification:** Clear value proposition

Areas of Excellence: You've effectively identified: 1. The subjective nature of current methods (memory, inconsistent photos) 2. Accessibility issues with professional dermatological tools 3. Limitations of existing mobile applications (reminders vs. tracking) 4. The opportunity for an objective, automated solution

Recommendations for Enhancement:

While your problem description is very strong, consider adding:

1. **User Research Evidence:**
   - If available, cite studies on consumer dissatisfaction with skincare products
   - Statistics on product return rates or abandonment
   - Example: "Studies indicate that X% of consumers abandon skincare routines within 4 weeks due to perceived lack of results" (if such data exists)
2. **Specific Examples of Existing App Limitations:**
   - Briefly mention 1-2 existing apps by name (e.g., SkinVision, MDacne) and their limitations
   - This would strengthen your competitive analysis
3. **Ethical/Privacy Consideration Preview:**
   - Brief mention that facial data will be handled sensitively
   - This shows awareness of potential concerns

Grade Potential for this section: 70-80%

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3. INTENDED OUTCOMES EXCELLENT

Assessment: This is the **strongest section** of your proposal.

Outstanding Elements: - Comprehensive technical deliverables clearly listed - Specific technologies named (OpenCV, PyTorch) - Full pipeline covered (capture → processing → analysis → visualization) - Security considerations mentioned (secure storage) - Ethical awareness (non-medical interpretation messages) - Testing and evaluation planned (accuracy across varied conditions) - Realistic scope (tested prototype, not commercial product)

Technical Pipeline Coverage: Your intended outcomes cover the complete system: 1. **Input:** Guided image capture, consistent positioning 2. **Processing:** Face detection, alignment, lighting normalization 3. **Analysis:** Feature extraction, detection algorithms, comparison 4. **Output:** Dashboard, visualizations, progress metrics 5. **Quality Assurance:** Testing across varied conditions and users

Particularly Commendable: - **"Clear non-medical interpretation messages"** - Shows awareness of regulatory/ethical boundaries. This is crucial for health-related applications. - **"Evaluation tests measuring accuracy"** - Demonstrates commitment to rigorous validation - **"Tested prototype"** - Realistic scope management

Recommendations for Enhancement:

1. **Quantify Testing Metrics:** Add specific success criteria:
   - "Achieve face detection accuracy of >95% under varied lighting"
   - "Track 5-10 volunteer users over 4-6 weeks"
   - "User satisfaction rating of 4/5 or higher"
   - "Algorithm ability to detect acne count changes with ±10% accuracy"
2. **Define Evaluation Methodology:**
   - How many test subjects? (Suggest 5-10 volunteers)
   - Duration of study? (Suggest minimum 4 weeks to see skin changes)
   - Ground truth comparison? (Manual counting vs. automated detection)
   - User experience evaluation? (System Usability Scale questionnaire?)
3. **Platform Specification:**
   - Will this be a mobile app (iOS/Android), web app, or desktop application?
   - You mention Flutter in references - clarify if this is your chosen platform

Action Item: Add specific evaluation metrics and testing protocols to AT2 report.

Grade Potential for this section: 75-85%

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4. CHALLENGE SUITABILITY VERY GOOD

Strengths: - Explicitly references COM668 learning outcomes - Mentions BCS accreditation guidelines - Lists relevant technical domains (computer vision, ML, software engineering) - Acknowledges independent self-management requirement - Highlights societal value

Good Coverage: You've identified integration of: - Computer vision techniques - Machine learning models - Software engineering practices - System design principles

Recommendations for Enhancement:

To achieve an **Excellent** grade in AT2, expand this section with:

1. Explicit Learning Outcomes Mapping:

Map your project to the four module learning outcomes:

- **LO1 (Synthesise and Evaluate Solutions):**
  - "Will synthesize computer vision algorithms, ML models, and UX design to create an innovative skincare tracking solution"
  - "Will critically evaluate effectiveness through quantitative accuracy metrics and user satisfaction surveys"
- **LO2 (Needs, Risks, and Societal Context):**
  - "Addresses consumer need for objective skincare assessment"
  - "Manages privacy risks through local data storage and encryption"
  - "Considers ethical boundaries (non-medical disclaimers)"
  - "Societal impact: democratizing access to objective skin analysis"
- **LO3 (Methodological Integration and Self-Management):**
  - "Will apply Agile methodology with 2-week sprints"
  - "Integration of analytical skills (algorithm design) and practical skills (implementation, testing)"
  - "Self-managed milestones: prototype → core features → testing → refinement"
- **LO4 (Professional Communication):**
  - "Technical documentation of algorithms and data structures"
  - "User-facing documentation and privacy policies"
  - "Critical reflection on technical decisions and limitations"

2. Technical Challenge Depth:

Articulate the significant technical challenges: - **Face alignment under varying poses:** Handling different angles, distances - **Lighting normalization:** Ensuring consistency across different environments - **Feature extraction accuracy:** Distinguishing acne from shadows, freckles, etc. - **Temporal tracking:** Matching facial regions across time with skin changes - **Real-time performance:** Processing speed on consumer devices - **Data privacy:** Secure local storage without cloud dependency (or encrypted cloud)

3. BCS Level 6 Specific Criteria:

Explicitly state how your project meets: - Substantial scope (400 hours): Multi-component system with research, implementation, testing - Technical complexity: AI/ML integration with computer vision - Professional practice: Following software development lifecycle, version control, testing - Critical evaluation: Systematic testing and user studies - Independent research: State-of-the-art CV/ML techniques

Action Item: Expand this section in AT2 Challenge Definition Report with explicit LO and BCS mapping.

Grade Potential for this section: 70-75% (can reach 80-85% with expansion)

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5. COPYRIGHT/IPR/COMMERCIAL SENSITIVITY EXCELLENT

Assessment: This section demonstrates **professional awareness** of intellectual property considerations.

Strengths: - Clear statement of open-source framework usage - Explicit claim of ownership for original work - Specific components listed (code, algorithms, scoring metrics, workflow) - Acknowledgment of no proprietary datasets - Recognition of no commercial restrictions

This is exactly the level of detail required.

Minor Enhancement Suggestion:

Consider adding a brief statement about: 1. **User data ownership:** - “Users retain ownership of their facial images and progress data” - “All data remains locally stored (or: encrypted and anonymized if cloud-based)”

2. **License for your work:**
   - Will you release this as open-source after submission? (Optional)
   - “Project code may be made available under MIT License post-assessment” (if applicable)
3. **Ethical Approval:**
   - “Will obtain Ulster University Ethics approval before user testing”
   - “Participants will provide informed consent for facial image capture”

Action Item: Minor additions suggested above (optional but strengthens section).

Grade Potential for this section: 80-85%

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6. RESOURCES REQUIRED VERY GOOD

Strengths: - Comprehensive technology stack listed - Specific version requirements (Python 3.9+) - All tools identified as free/open-source - Data sources identified (Kaggle, volunteers, academic datasets) - Ethical consideration (informed consent) - Timeline feasibility acknowledged

Technology Stack Coverage: - **Languages:** Python (appropriate choice) - **CV/ML Libraries:** OpenCV, TensorFlow/Keras (Note: PyTorch in references but TensorFlow here - see note below) - **Backend:** Flask (good for prototype) - **Database:** SQLite (appropriate for local storage) - **Development Tools:** Git, VS Code (professional choices) - **Hardware:** Standard laptop, webcam/smartphone (realistic)

Important Clarification Needed:

TensorFlow vs. PyTorch: - Your **Intended Outcomes** mention **PyTorch** - Your **Resources** mention **TensorFlow/Keras** - Your **References** include PyTorch documentation

Action Required: Choose one framework and be consistent throughout your project: - **PyTorch:** More flexible, better for research, steeper learning curve - **TensorFlow/Keras:** More production-ready, easier to deploy, simpler API

Recommendation: Given you're doing custom skin feature detection, **PyTorch** might offer more flexibility. Make this decision early and update all documentation consistently.

Additional Recommendations:

1. **Frontend Platform Clarity:**
   - You mention Flutter in references but not in resources
   - **Clarify:** Mobile app (Flutter/React Native) or Web app (Flask + HTML/CSS/JS) or Desktop (Python GUI)?
   - **Suggestion:** For a 400-hour project, consider web-based interface (simpler deployment) unless mobile is essential
2. **Testing Environment:** Add:
   - Various lighting conditions (daylight, indoor, evening)
   - Different camera qualities (high-end smartphone vs. webcam)
   - Diverse skin tones (ensure algorithm fairness)
3. **Data Volume Estimates:**
   - How many images per user? (e.g., daily captures over 6 weeks = 42 images)
   - Storage requirements? (1000x1000px images = ~500KB each = 21MB per user)
4. **Computing Resources:**
   - Will you need GPU for model training? (Your laptop specifications?)
   - Pre-trained models vs. training from scratch? (Recommend pre-trained for time efficiency)

Action Item: Clarify TensorFlow vs. PyTorch and specify frontend platform.

Grade Potential for this section: 70-75% (can reach 80% with clarifications)

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7. REFERENCES OUTSTANDING

Assessment: This is an **exemplary** reference list for a project proposal.

Strengths: - **10 high-quality references** (well above minimum) - **Mix of source types:** Academic papers, books, technical documentation - **Proper Harvard formatting** throughout - **Relevant and current:** 2016-2026 timeframe - **Authoritative sources:** IEEE, MIT Press, academic journals - **Technical depth:** Covers theory and practical implementation

Reference Coverage Analysis:

Category	Count	Examples
Technical Documentation	3	OpenCV, PyTorch, Flutter docs
Foundational ML/CV Books	2	Goodfellow (Deep Learning), Szeliski (Computer Vision)
Academic Papers	4	Zhang et al. (Face Detection), Kollias (Skin Analysis), Ojala (Texture Analysis)
Research Libraries	2	Bradski (OpenCV), Paszke (PyTorch)

Particularly Strong Citations:

1. **Zhang et al. (2016) - Multitask Cascaded CNNs:** Directly relevant to face detection/alignment
2. **Ojala et al. (2002) - Local Binary Patterns:** Excellent for texture analysis (skin texture tracking)
3. **Kollias & Baqer (1987) - Melanin spectroscopy:** Shows depth of research into skin analysis
4. **Szeliski (2022) - Computer Vision:** Recent edition, comprehensive foundation

This demonstrates you've done thorough background research beyond just technical documentation.

Minor Recommendations for AT2:

1. **Add Application-Specific References:**
   - Research on existing skincare tracking apps (even if limited)
   - Studies on consumer behavior with skincare products
   - Medical/dermatology papers on acne assessment methods (for validation)
2. **Machine Learning Architecture Papers:**
   - Specific papers on CNN architectures for skin lesion detection
   - Transfer learning approaches (e.g., ResNet, MobileNet papers)
3. **User Experience Research:**
   - Studies on health app usability

– Privacy concerns in biometric applications

Example Additional References for AT2:

```
Esteva, A. et al. (2017) 'Dermatologist-level classification of skin cancer
with deep neural networks', Nature, 542(7639), pp. 115-118.

He, K. et al. (2016) 'Deep Residual Learning for Image Recognition',
in Proceedings of IEEE Conference on Computer Vision and Pattern Recognition,
pp. 770-778.
```

Action Item: Expand reference list in AT2 to 15-20 sources covering additional areas above.

Grade Potential for this section: 85-90% (already excellent, can reach 90%+ with expansions)

TECHNICAL DEEP-DIVE & RECOMMENDATIONS

Architecture Considerations

Current Approach (Inferred): 1. Image capture with guided positioning 2. Face detection → alignment → normalization 3. Feature extraction using ML models 4. Comparison against historical baseline 5. Visualization and scoring

Recommendations:

1. Face Detection & Alignment

Good Choice: Zhang et al.'s MTCNN (referenced) is excellent for this.

Implementation Path: - Use pre-trained MTCNN for face detection - Extract facial landmarks (68-point or 5-point model) - Apply affine transformation for alignment - Normalize to standard size (e.g., 224×224 for ML models)

Challenge to Address: - What if face is partially occluded (hand touching face, hair)? - Handling different distances from camera (zoom variations)? - **Solution:** Implement quality checks and guide user to recapture

2. Lighting Normalization

Critical for Consistency: - Different lighting drastically affects perceived skin color/texture - Must normalize to enable accurate comparison

Suggested Approaches: - **Histogram Equalization:** Basic but effective - **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Better for local contrast - **Color Constancy Algorithms:** Gray World, White Patch - **Reference Object:** Ask user to include color reference card in frame (optional, more complex)

Recommendation: Start with CLAHE (available in OpenCV), evaluate effectiveness, enhance if needed.

3. Skin Feature Detection

Three Target Metrics: Acne, Redness, Texture

A. Acne Detection

Approach 1 - Traditional CV: - Use blob detection (OpenCV SimpleBlobDetector) - Morphological operations to identify raised/dark spots - **Pros:** Fast, interpretable - **Cons:** Less accurate, sensitive to lighting

Approach 2 - Deep Learning: - Fine-tune pre-trained CNN (ResNet, MobileNet) on acne dataset - Train binary classifier: acne spot vs. clear skin - Use sliding window or object detection (YOLO, Faster R-CNN) - **Pros:** More accurate - **Cons:** Requires training data, more complex

Recommendation: Start with **Approach 1** for MVP, upgrade to **Approach 2** if time permits and you find good training data.

B. Redness Detection

Approach: - Convert to HSV color space - Define red hue range (H: 0-10, 160-180) - Calculate percentage of skin pixels in red range - Track changes over time

Challenge: Distinguish inflammation redness from natural skin tone variations **Solution:** Establish baseline from first few captures, track relative changes

C. Texture Analysis

Excellent Reference: Your citation of Ojala et al. (LBP - Local Binary Patterns) is perfect for this.

Approach: - Apply LBP to skin regions - Calculate texture histograms - Smoothness score: Lower LBP variance = smoother skin - Track changes over time

Alternative: Compute standard deviation of pixel intensities in skin regions

4. Temporal Tracking & Comparison

Key Challenge: Matching same skin regions across different captures

Approach: 1. Use facial landmarks to define regions of interest (ROIs): - Forehead region - Left cheek, right cheek - Nose, chin 2. Extract and align each ROI independently 3. Apply feature detection to each ROI 4. Store per-ROI metrics over time 5. Compute per-ROI trend scores

Visualization: - Line graphs showing metric changes over time - Side-by-side before/after comparisons with aligned overlays - Heatmaps showing improvement areas (green) vs. worsening areas (red)

5. Scoring & Metrics

Recommendation: Develop a composite "Progress Score"

```
Progress Score = (Acne_Improvement × 0.4) +
                  (Redness_Reduction × 0.3) +
                  (Texture_Smoothness × 0.3)

Where each component is normalized to 0-100 scale:
- 0 = worsening from baseline
- 50 = no change
- 100 = maximum improvement
```

Present to users as: - Overall progress percentage - Per-metric breakdown - Trend direction (improving/stable/worsening)

6. Platform Architecture

Given your tech stack, recommended architecture:

Option A: Web Application - **Frontend:** HTML/CSS/JavaScript (or React for richer UI) - **Backend:** Flask API (Python) - **Processing:** OpenCV + PyTorch on server - **Storage:** SQLite locally or PostgreSQL for multi-user - **Deployment:** Local or Heroku/PythonAnywhere for testing

Option B: Mobile Application - **Frontend:** Flutter (cross-platform iOS/Android) - **Backend:** Flask API or Firebase - **Processing:** Server-side (heavy ML) + some client-side (face detection) - **Storage:** Local SQLite + optional cloud backup

Recommendation for 400-hour scope: Web application (Option A) is more achievable and allows focus on core CV/ML algorithms rather than mobile development complexities. You can still make it mobile-responsive.

If you choose mobile: Consider hybrid approach where heavy processing happens server-side via API calls.

ETHICAL & REGULATORY CONSIDERATIONS

Critical Points to Address

1. Medical Device Regulations

Important: Your app analyzes skin conditions (acne, redness)

Risk: Could be classified as medical device in some jurisdictions

Mitigation (You've already started this): - "Clear non-medical interpretation messages" (mentioned in Intended Outcomes) - Emphasize it's a "tracking tool" not a "diagnostic tool" - Include disclaimers: "This app does not diagnose, treat, or provide medical advice. Consult a dermatologist for medical concerns." - Focus messaging on "skincare product effectiveness" not "skin condition diagnosis"

Action Item: Ensure all user-facing text clearly states non-medical purpose.

2. Data Privacy (GDPR Compliance)

Facial images are biometric data = special category under GDPR

Requirements: - Explicit informed consent before capturing images - Clear privacy policy explaining data usage - Right to deletion (users can delete their data) - Secure storage (encryption recommended) - No third-party sharing without explicit consent - Minimal data collection (only capture what's needed)

Recommendation: - **Local storage first:** Keep all data on user's device (simplest, most private) - **If cloud storage:** Encrypt before upload, anonymize identifiers - **For testing:** Anonymize volunteer data, delete after project completion

Action Item: Ethics application will address these - work with mentor on this.

3. Algorithmic Fairness

Risk: CV/ML models can perform differently across skin tones

Historical Issue: Many face detection models trained predominantly on lighter skin, perform worse on darker skin tones

Mitigation: - Test with diverse skin tones (Fitzpatrick scale I-VI) - Use diverse training data if training custom models - Report accuracy metrics segmented by skin tone - Acknowledge limitations if fairness gaps exist

Action Item: Include diversity testing in evaluation plan (AT2 & AT4).

4. User Safety

Potential Risks: - User makes skincare decisions based on potentially inaccurate algorithm - User delays seeking professional help for serious skin condition - Algorithm misclassifies something serious as benign

Mitigation: - Clear disclaimers and limitations - Encourage professional consultation for concerns - Honest reporting of algorithm accuracy/limitations in reports

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METHODOLOGY RECOMMENDATIONS

For AT2 Challenge Definition Report, you'll need to justify your chosen software development methodology.

Recommended Approach: Agile (Scrum-based)

Rationale: 1. **Iterative Development:** Build MVP, test, refine algorithms based on results 2. **Flexible Requirements:** Algorithm parameters may need tuning based on testing 3. **Regular Deliverables:** Sprint demos align with AT2, AT3, AT4 milestones 4. **Risk Management:** Early prototyping reduces technical risk

Suggested Sprint Structure (2-week sprints):

Sprint 1-2 (Weeks 1-4): Research & Basic Prototype - Literature review and technology evaluation - Basic face detection and alignment implementation - Test with sample images

Sprint 3-4 (Weeks 5-8): Core Feature Development - Implement lighting normalization - Develop acne detection algorithm - Redness and texture analysis - AT2 Challenge Definition Report preparation

Sprint 5-6 (Weeks 9-12): Feature Integration & Storage - Historical comparison logic - Database schema and implementation - Basic UI/dashboard development - AT2 submission (Week 11)

Sprint 7-8 (Weeks 13-16): Testing & Refinement - Algorithm accuracy testing - User testing with volunteers (recruit early!) - Bug fixes and performance optimization

Sprint 9-10 (Weeks 17-20): Advanced Features & Evaluation - Visualization dashboard enhancements - Comprehensive evaluation study - AT3 Demo Video preparation - AT3 submission (Week 18)

Sprint 11-12 (Weeks 21-24): Final Testing & Documentation - Final user study - Performance benchmarking - AT4 Project Review Report - AT4 submission (Week 23)

Alternative: Incremental/Spiral Model also defensible given the research + development nature.

Action Item: Choose and justify methodology in AT2 report.

RISK ANALYSIS

For AT2, you'll need a comprehensive risk analysis. Here are key risks to address:

Risk	Likelihood	Impact	Mitigation Strategy
Face detection fails in poor lighting	High	Medium	Implement quality checks, guide user to retake
Algorithm accuracy	Medium	High	Start with simple baseline, iteratively improve; set realistic accuracy targets

insufficient			
Insufficient training data	Medium	Medium	Use pre-trained models, augment with Kaggle datasets, synthetic data
Processing too slow for user tolerance	Medium	Medium	Optimize algorithms, use efficient models (MobileNet), process asynchronously
Can't recruit enough test volunteers	Medium	High	Start recruitment early, use snowball sampling, offer small incentives (if allowed)
Skin changes too subtle over project timeline	High	High	Use synthetic test cases with known differences, shorter testing period (2-3 weeks), accept limitations
Scope creep (too many features)	Medium	High	Strict prioritization: Must/Should/Could/Won't; focus on core tracking features
Platform compatibility issues	Low	Medium	Test on multiple devices early, choose widely supported technologies
Privacy/ethics approval delays	Medium	High	Submit ethics application early (AT1 stage), have backup plan using only synthetic/stock images

Action Item: Expand and detail this table in AT2 Risk Analysis section.

EVALUATION STRATEGY

For AT3 (Demo) and AT4 (Review), you'll need robust evaluation. Plan now:

Quantitative Metrics

1. Algorithm Accuracy: - **Ground Truth:** Manual counting/measurement by you (or dermatologist if accessible) - **Test Dataset:** 50-100 face images with varied acne levels, lighting, skin tones - **Metrics:** - Acne detection precision/recall/F1-score - Redness correlation with manual assessment (Pearson's r) - Texture score correlation with dermatologist ratings

Target: >80% accuracy for acne detection, >0.7 correlation for redness/texture

2. System Performance: - Processing time per image (target: <10 seconds) - Memory usage - Success rate of face detection (target: >95%)

3. User Study: - **Participants:** 5-10 volunteers - **Duration:** 4-6 weeks with 2-3 captures per week - **Measures:** - Task completion rate - System Usability Scale (SUS) score (target: >70) - User-reported satisfaction (5-point Likert scale) - Perceived usefulness vs. manual photo comparison

Qualitative Evaluation

4. User Feedback: - Post-study interviews or surveys - Perceived accuracy of tracking - Usefulness for decision-making - Suggested improvements

5. Critical Reflection: - What worked well technically? - What were the limitations? - How did reality differ from plan? - What would you do differently?

Action Item: Detail evaluation methodology in AT2 Planning section.

TIMELINE & MILESTONES

Key Dates (from Module Handbook):

Deadline	Assessment Task	Your Focus
Oct 9, 2026	AT1 Proposal (This)	Approved (subject to ethics)
Oct 23, 2026	Ethics Approval	Mentor submits; you support with documentation
Dec 4, 2026	**AT2 Challenge Definition**	Research, design, initial prototype, risk analysis
Mar 5, 2027	**AT3 Demo Video + Code**	Working prototype, feature demonstrations, code walkthrough

Apr 23, 2027	**AT4 Project Review**	Evaluation results, critical appraisal, reflection

Recommended Internal Milestones:

- **Week 4:** Basic face detection working
- **Week 6:** One feature (acne detection) functional
- **Week 8:** All three features (acne/redness/texture) implemented
- **Week 10:** AT2 report complete
- **Week 14:** Historical comparison and storage working
- **Week 16:** UI/dashboard functional
- **Week 18:** User testing begins
- **Week 20:** AT3 demo recorded
- **Week 22:** Evaluation complete
- **Week 24:** AT4 report complete

Action Item: Create detailed Gantt chart for AT2 Planning section.

ASSESSMENT RUBRIC PROJECTIONS

Based on AT2 Challenge Definition Report rubric:

Criterion	Weight	Current Proposal Quality	Projected AT2 Grade with Recommendations
Problem Definition	20%	75-80%	75-85%
Context Investigation	20%	80-85%	80-90%
Software Definition	40%	70-75%	75-85%
Planning	15%	65-70%	75-85%
Communication	5%	80-85%	80-85%

Overall AT2 Projection: 70-85% (Excellent range) if you follow through on recommendations

AT3 & AT4 Potential: With strong implementation and thorough evaluation, **75-90% achievable**

ACTION ITEMS SUMMARY

Priority 1 - For AT2 Challenge Definition Report (Critical):

1. **Expand Challenge Suitability section** with explicit LO1-4 and BCS criteria mapping
2. **Clarify technology choice:** PyTorch vs. TensorFlow/Keras - be consistent
3. **Specify platform:** Web app, mobile app, or desktop? Justify choice
4. **Define evaluation metrics:** Specific accuracy targets, user study protocol
5. **Comprehensive risk analysis:** Use table provided as starting point
6. **Justify methodology:** Agile vs. other approaches
7. **Create project timeline:** Gantt chart with milestones

Priority 2 - Before Implementation Begins (Important):

8. **Ethics approval preparation:** Work with mentor on application
9. **Recruit test volunteers early:** Need 5-10 participants with 4-6 week commitment
10. **Decide on architecture:** Finalize technical architecture diagram
11. **Set up development environment:** Version control, testing framework
12. **Expand references:** Add 5-10 more for AT2 (application-specific, ML architectures, UX)

Priority 3 - During Development (Ongoing):

13. **Algorithm selection research:** Benchmark different approaches for each feature
14. **Fairness testing:** Ensure diverse skin tone testing
15. **Performance optimization:** Meet <10 second processing target
16. **Documentation:** Keep detailed notes for AT4 reflection

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SPECIFIC RECOMMENDATIONS FOR EXCELLENCE (70-85% TARGET)

To move from “Very Good” to “Excellent” grade:

1. Technical Innovation

- Don’t just implement standard algorithms - **adapt and enhance** them
- Example: Combine multiple texture metrics (LBP + GLCM + fractal dimension) and compare effectiveness
- Document your experimental process and comparative analysis

2. Rigorous Evaluation

- **Comparative study:** Your algorithm vs. manual assessment vs. baseline (simple difference)
- **Statistical analysis:** Use appropriate tests (t-tests, correlation, ANOVA)
- **Failure analysis:** Don’t just report successes - deeply analyze failure cases

3. Professional Documentation

- **Architecture diagrams:** Clear system architecture, data flow, algorithm flowcharts
- **Code quality:** Clean, well-commented, follows PEP 8 (for Python)
- **Version control:** Meaningful commit messages, branching strategy
- **Testing:** Unit tests for core functions, integration tests for pipeline

4. Critical Reflection

- **Honest limitations:** Acknowledge what didn't work and why
- **Alternative approaches:** Discuss what you would do differently
- **Future work:** Concrete suggestions for improvements (not vague "add more features")

5. Presentation Quality

- **AT3 Demo:** Rehearse multiple times, show both successes and limitations
- **Visualizations:** Professional-quality charts and comparisons
- **User-centered narrative:** Show how it helps real users

COMPARISON: SIMILAR PROJECTS & DIFFERENTIATION

Existing Apps/Research to Acknowledge:

1. **SkinVision:** Commercial app for skin cancer detection
   - Your differentiation: Focus on skincare progress, not medical diagnosis
2. **MDacne:** Acne tracking with photos
   - Your differentiation: Automated alignment and objective metrics, not just photo gallery
3. **HiMirror:** Smart mirror with skin analysis
   - Your differentiation: Software-only solution, more accessible/affordable
4. **Academic Research:** Multiple papers on automated acne detection
   - Your differentiation: Complete tracking system with UI, not just algorithm paper

Your Unique Value: - Consumer-focused (not medical) - Automated consistency (alignment, lighting) - Multi-metric tracking (acne + redness + texture) - Temporal analysis with quantified progress - Affordable/accessible (no special hardware)

Action Item: Include comparative table in AT2 Context Investigation section.

POTENTIAL CHALLENGES & SOLUTIONS

Challenge 1: "Skin changes are slow - won't see results in 4 weeks"

Solution: - Use synthetic test pairs: Take photos, artificially add/remove spots, test detection - Shorter-cycle testing: Makeup application/removal as rapid test - Accept limitation: Document that real-world validation needs longer timeline - Compare progress tracking accuracy, not actual skin improvement

Challenge 2: "ML model needs huge training data"

Solution: - Use transfer learning: Pre-trained models (ResNet, MobileNet) on ImageNet - Fine-tune on small dermatology datasets (Kaggle has some) - Consider traditional CV for MVP, ML as enhancement if time permits - Clearly state in reports if using pre-trained vs. custom training

Challenge 3: "Users forget to take daily photos"

Solution: - Design for flexibility: Works with any frequency (daily, weekly) - Focus on technical accuracy, not user compliance (out of scope) - Recommendation system: "Based on your usage pattern…"

SUPERVISOR ASSESSMENT

Strengths of This Proposal (Exceptional):

Clear, measurable, ambitious aim
Well-researched problem with market validation
Comprehensive technical approach
Excellent reference list showing depth of research
Realistic scope with clear deliverables
Awareness of ethical and privacy considerations
Strong alignment with module requirements
Professional presentation and communication

Areas Requiring Clarification (Minor):

Technology consistency (PyTorch vs. TensorFlow)
Platform specification (Web vs. Mobile)
Evaluation metrics (need quantification)
Challenge suitability (needs expansion for AT2)

Overall Recommendation:

This is one of the strongest project proposals I've reviewed. It demonstrates: - **Technical ambition balanced with realistic scope - Clear understanding of the problem domain - Thorough background research - Professional approach to planning**

The project has excellent potential for an Excellent or Professional grade (70-90%) if you: 1. Maintain the rigorous approach shown in this proposal 2. Address the technical clarifications needed 3. Execute thorough evaluation and testing 4. Provide critical, honest reflection in AT4

NEXT STEPS

Immediate Actions (This Week):

1. **Acknowledge this feedback** and confirm understanding
2. **Make technology decisions:** PyTorch vs. TF, Web vs. Mobile
3. **Ethics preparation:** Draft consent forms and data handling protocol
4. **Set up development environment:** Git repo, IDE, initial dependencies

Short-term (Next 2-4 Weeks):

5. **Begin implementation:** Start with face detection module
6. **Literature deep-dive:** Read referenced papers in detail
7. **Recruit volunteers:** Start identifying test participants
8. **AT2 preparation:** Begin drafting sections as you research/design

Ongoing:

9. **Weekly progress reviews:** Schedule regular check-ins with me
10. **Documentation:** Keep detailed research and development log
11. **Version control:** Commit regularly with meaningful messages

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RESOURCES & SUPPORT

Technical Resources:

OpenCV Tutorials: - Face detection: https://docs.opencv.org/4.x/d2/d99/tutorial_js_face_detection.html - Image alignment: https://learnopencv.com/image-alignment-feature-based-using-opencv-c-python/

PyTorch Resources: - Transfer learning: https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html - Computer vision: https://pytorch.org/vision/stable/index.html

Skin Analysis Papers: - Search "automated acne detection" on Google Scholar - IEEE Xplore: skin lesion detection papers

Ulster University Support:

- **Library:** Subject librarian for Computing - can help find relevant papers
- **Ethics:** Your mentor will guide ethics application process
- **Writing Support:** Student Success Centre for report writing
- **Studiosity:** 24-hour draft feedback service

My Availability:

- **Office Hours:** [Specify your times]
- **Email Response:** Within 24-48 hours

- **Meeting Requests:** Book via [method]

FINAL COMMENTS

Sanju, this is an **exemplary project proposal** that sets a high standard. Your proposal demonstrates:

Strong research skills - Evident from your reference list and problem analysis
Technical competence - Appropriate technology choices and clear understanding
Professional communication - Well-structured, clear, comprehensive
Realistic planning - Ambitious yet achievable scope
Ethical awareness - Recognition of privacy and medical boundaries

You should be confident in your ability to deliver an excellent project. The foundation you've laid in this proposal is solid. Now the focus is on: 1. **Execution:** Implementing your planned system effectively 2. **Evaluation:** Rigorous testing and honest assessment 3. **Reflection:** Critical analysis of what worked and what didn't

My confidence in your success: Very High

The main risks are: - Scope creep (stay focused on core features) - Over-optimism on algorithm accuracy (set realistic targets) - Insufficient time for user testing (start recruiting early)

With careful execution, you're on track for a 75-85% grade, potentially higher.

I look forward to seeing your progress. Please don't hesitate to reach out with questions or for guidance as you move forward.

Proposal Status: APPROVED
Ethics Approval: Required - mentor will initiate
Next Submission: AT2 Challenge Definition Report - December 4, 2026

Supervisor Signature: Dr. Oguzie Goodluck
Date: February 12, 2026

Appendix: Suggested AT2 Outline Enhancement

When you write AT2, consider this enhanced structure:

1. Problem Definition (650 words) - Expand on market need with statistics - User personas (e.g., "Sarah, 22, international student…") - Consequences of current situation (financial, psychological)

2. Context Investigation (950 words) - Legal/ethical: GDPR, medical device regulations, informed consent - Theory: Computer vision techniques, ML architectures - Similar products: Detailed comparison table (see section above) - Expand references to 15-20

3. Software Definition (1500 words) - Requirements: Functional and non-functional, prioritized (MoSCoW) - Risk analysis: Expanded table with mitigation strategies - System design: Architecture diagram, data model, algorithm flowcharts - UI/UX design: Wireframes or mockups - Rationale: Why these choices?

4. Planning (500 words) - Methodology: Agile justification with sprint structure - Timeline: Gantt chart with milestones - Testing strategy: Unit, integration, user testing plans - Version control and documentation approach

Total: 3600 words (perfect for 45% assessment task)

End of Feedback Document

Good luck with your project, Sanju! I'm excited to see this develop.